

Pricing the Dormitory Book

A risk-based rating algorithm for ABG Dormitory Insurance

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What we found

4.5x

Greek members cost four to five times non-members, on every coverage

1.8x

Off-campus students cost nearly twice on-campus, frequency and severity compound

\$948

Average Gross Premium for Non-Greek life applicants

\$3,386

Average Gross Premium for Greek life applicants

Our recommendation

Price Greek and Non-Greek as two separate programs. Each carries its own base rates and its own factor tables, so the difference in risk is reflected in the rate rather than absorbed by the rest of the book.

What we were asked to build

Coverage	Limit	What it pays for
Personal Property	\$10,000	Theft or damage to belongings
Additional Living Expense	Not stated	Costs when the dorm is uninhabitable
Guest Medical	\$150,000	Medical costs for injured guests, no fault
Liability	\$500,000	Defence and damages if a guest sues

The deliverable

- A stand-alone rating algorithm per coverage
- A base rate per rating tier
- Factors applied within each tier

10,000 policyholders · 1,819 claims · one policy year · four coverages on every policy

Every student carries all four coverages, so the policy premium is the sum of four separately rated pieces.

We verified the data before we modelled it



No deductible

Smallest claims are \$27, \$221, \$275, \$316 — no round floor, no gap at zero



No limit censoring

Largest Liability claim is \$42,915 against a \$500,000 limit; no mass at any limit



Single period, clean

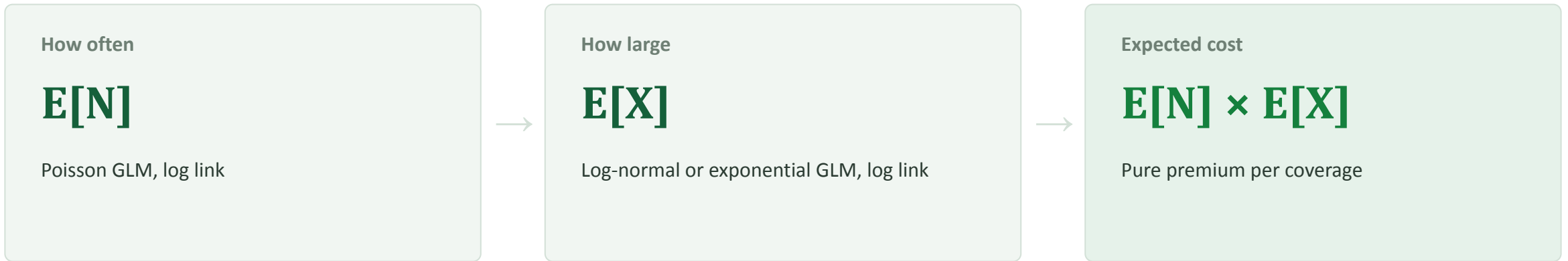
One policy year, no missing values, every student attribute constant



Exposure is uniform

Exactly one policy-year each, so no offset is needed in the frequency models

We priced frequency and severity separately



Why not model loss cost directly

- The response is mostly zeros with a long right tail, which no standard GLM family describes well. Counts and claim sizes each have families that fit them properly.
- A loss cost per exposure is the product of two different processes. One model fitted to the product cannot say whether a group is expensive because it claims more often or because its claims are larger.
- When a variable pushes frequency one way and severity the other, a combined model sees the two effects cancel and drops a variable that genuinely matters.
- Two models with log links give two sets of relativities that multiply into a rate table a regulator can read line by line.

Frequency: claim counts are Poisson on every coverage

Coverage	Frequency	Variance / mean	Selected	Δ BIC vs next
Personal Property	0.097	1.014	Poisson	8.5
Additional Living Expense	0.053	1.008	Poisson	8.6
Guest Medical	0.023	1.011	Poisson	8.4
Liability	0.009	0.992	Poisson	8.7

How we got there

- Variance-to-mean ratio as the first screen. A ratio at 1.00 is the Poisson fingerprint; above 1 points to negative binomial or zero inflation, below 1 to binomial.
- Fitted Poisson, negative binomial, ZIP and ZINB by maximum likelihood and ranked on BIC. Zero-inflated forms needed direct numerical optimisation since they are not GLM families.

Severity: claim sizes are log-normal, except Liability

Coverage	Claims	Mean severity	Selected	Δ BIC vs next
Personal Property	787	\$1,023	Log-normal	31.0
Additional Living Expense	427	\$5,514	Log-normal	15.6
Guest Medical	187	\$16,612	Log-normal	3.2
Liability	85	\$8,675	Exponential	2.28

Candidates tested

Log-normal · gamma · Weibull · exponential · inverse Gaussian, fitted by maximum likelihood on each coverage and ranked on BIC.

Why exponential for Liability

Exponential ranked best on BIC among the candidates tested, by 2.28 over the next distribution. With one parameter rather than two it is also the least demanding form to estimate on 85 claims.

How we chose which variables to price on

1

Split before fitting

Roughly 8,000 students to train, 2,000 held out and never touched during selection.

2

Forward stepwise

Add the variable that most reduces out-of-sample deviance; stop when nothing improves it.

3

A model per coverage

Frequency and severity were selected independently on each of the four coverages, so a variable earns its place in one model without being forced into the others

4

Collinearity check

Off-campus against distance to campus: $r = 0.642$, $VIF = 1.70$. Low enough to keep both.

*Excluded by judgment:
name and student ID carry no meaning
risk tier is an underwriting output, not something an applicant supplies.*

The selected predictors, by coverage

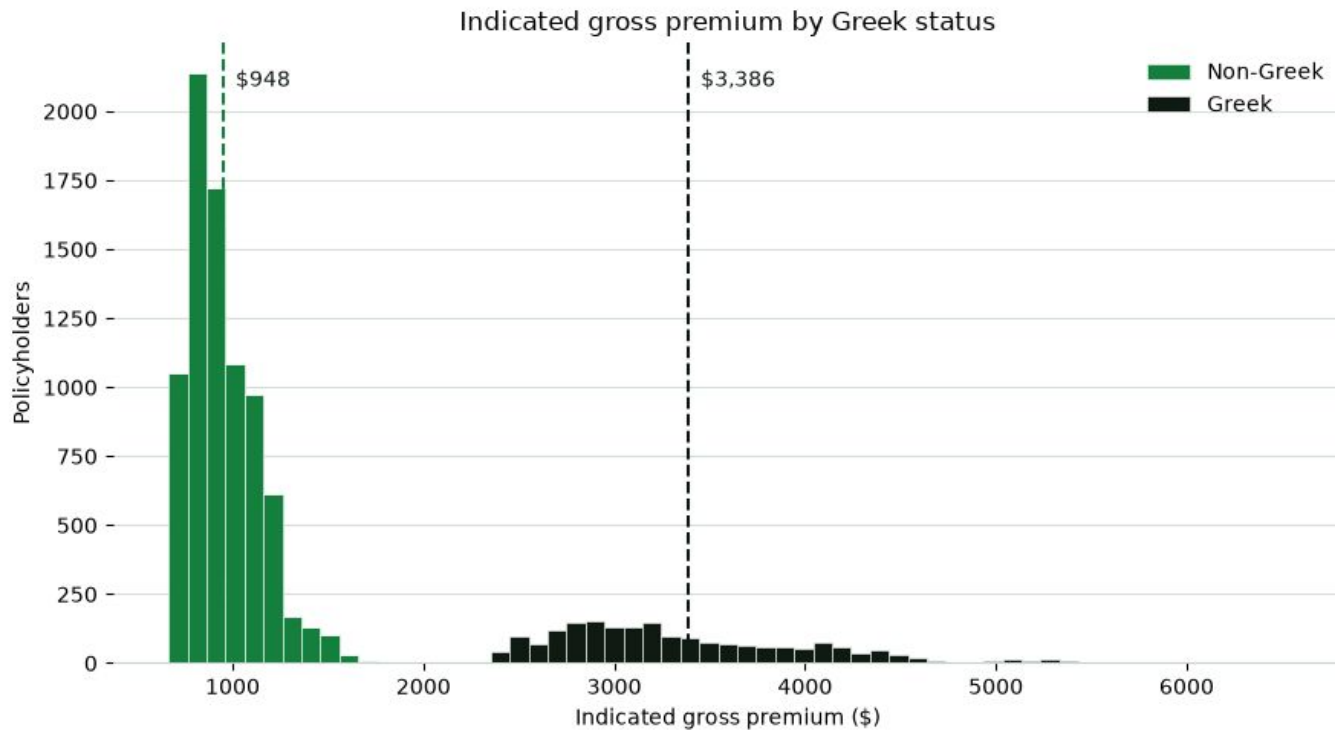
Coverage	Frequency model	Severity model
Personal Property	Greek, sprinklered, GPA	Greek, off-campus, sprinklered, study, distance
Additional Living Expense	Greek, sprinklered	Greek, off-campus, class year
Guest Medical	Greek, GPA, study	Greek, class year
Liability	Greek, distance, GPA, sprinklered, off-campus	Distance to campus

Greek membership is the only variable selected in all eight models.

Class year appears only in ALE and Guest Medical severity, so our tier dimension differentiates two of the four coverages. Distance to campus survives on Liability despite being related to off-campus, and we treat it as provisional.

KEY FINDING

The book splits into two populations



Why we split the book

- Two clear clusters with an empty middle
- Non-Greek averages \$948; Greek averages \$3,386.
- The book mean of \$1,427 falls in the gap. Almost nobody is priced at the average.
- Greek members are 19.6% of the book and are selected in all eight models.

Charging everyone the average would overcharge 80% of students and undercharge the rest.

Two programs, four tiers, priced per coverage

Premium = Base rate (program, tier, coverage) × factors within tier, summed across the four coverages

Tier is assigned once, at the policy level

Preferred-Plus

Undergraduate, on campus

Preferred

Undergraduate, off campus

Standard

Graduate, on campus

Non-Standard

Graduate, off campus

Factors applied inside the tier

- Sprinklered, GPA band, field of study — all four tiers
- Class year — undergraduate tiers only, since the tier already prices undergraduate against graduate
- Distance to campus — off-campus tiers only, since it is zero for every on-campus student

Why the tiers are cut this way

Mean indicated gross premium per policy

Non-Greek	On campus	Off campus
Undergraduate	826.70	1,064.27
Graduate	1,175.60	1,468.06

Greek	On campus	Off campus
Undergraduate	2,939.53	3,851.70
Graduate	4,260.14	5,389.59

What the four cells show

Both programs order the same way. Undergraduate on campus is the cheapest cell and graduate off campus the most expensive

The two dimensions move independently. Housing adds roughly 25 to 31% within each academic level, and academic level adds roughly 40% within each housing status, so neither is a proxy for the other.

Greek runs about 3.6 times Non-Greek in every one of the four cells, which is why program sits above the tier rather than inside it.

Preferred-Plus, Preferred, Standard and Non-Standard are these four cells, named.

Indicated base rates

Program	Tier	Personal Property	Additional Living Exp.	Guest Medical	Liability	Total policy
Non-Greek	Preferred-Plus	84.85	227.47	455.42	58.96	826.70
	Preferred	128.09	349.75	455.76	130.68	1,064.27
	Standard	84.91	325.21	705.69	59.80	1,175.60
	Non-Standard	128.94	506.82	698.30	133.99	1,468.06
Greek	Preferred-Plus	343.77	951.07	1,495.53	149.15	2,939.53
	Preferred	539.54	1,492.09	1,479.17	340.90	3,851.70
	Standard	346.71	1,399.89	2,363.24	150.30	4,260.14
	Non-Standard	511.37	2,197.88	2,335.51	344.83	5,389.59

Personal Property and Liability barely move between Preferred-Plus and Standard, because academic level is in neither model. Our tier dimension prices ALE and Guest Medical; housing prices the other two.

Factors applied within the tier

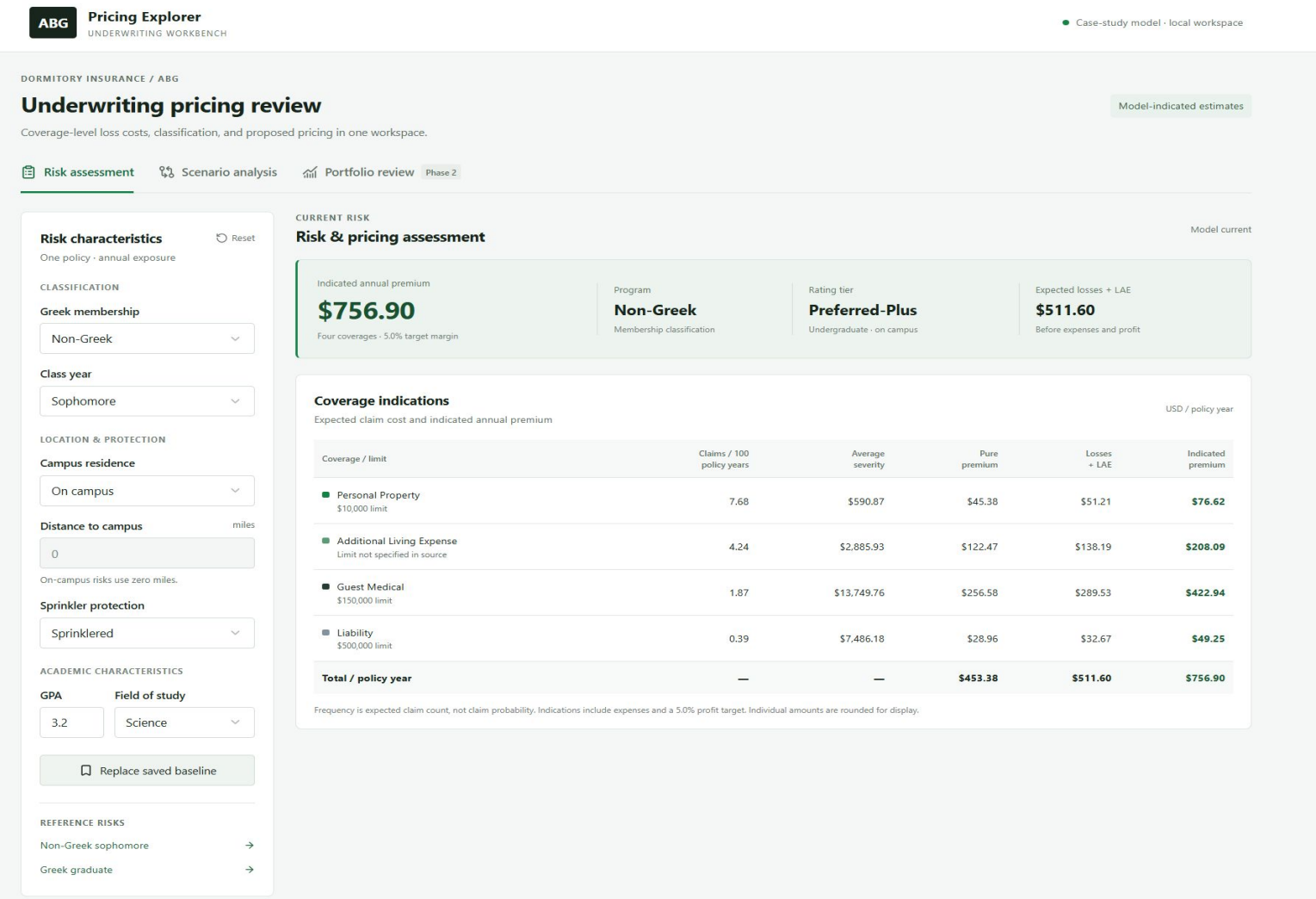
Non-Greek • Preferred (undergraduate, off campus) — one of eight such tables

Variable	Level	PP	ALE	GM	Liab.
Sprinklered	No	1.238	1.071	1.005	1.090
	Yes	0.899	0.970	0.998	0.962
GPA	0.00 – 0.99	1.100	1.000	1.071	1.247
	1.00 – 1.99	1.034	1.001	1.037	1.086
	2.00 – 2.99	0.986	0.996	0.987	1.005
	3.00 – 4.00	0.958	1.005	0.962	0.827
Field of study	Business	1.046	0.999	1.011	0.962
	Humanities	0.960	1.003	1.045	1.024
	Other	0.962	0.998	0.973	1.013
	Science	1.035	1.000	0.968	1.001

Variable	Level	PP	ALE	GM	Liab.
Class year	Freshman	1.001	0.908	0.838	0.998
	Sophomore	0.997	0.938	1.001	1.012
	Junior	1.004	1.092	1.167	0.980
	Senior	0.998	1.061	0.993	1.009
Distance	under 1 mile	0.996	1.004	1.001	0.661
	1 – 2 miles	0.995	0.999	1.003	0.731
	2 – 3 miles	0.993	1.002	0.999	0.798
	3 – 5 miles	1.005	0.999	0.997	0.918
	5 – 10 miles	1.006	0.999	1.001	1.245
	10 miles or more	1.006	0.991	0.995	2.435

PP = Personal Property, ALE = Additional Living Expense, GM = Guest Medical. Factors are normalised within the tier, so they average to 1.00. Distance carries almost all of its weight on Liability, where the farthest band is 2.435.

Underwriter pricing dashboard



- **Proposed pricing review:** Lets underwriters compare a proposed annual premium with the model indication and assess expected underwriting profit, margin, combined ratio, and break-even pricing.
- **Premium allocation:** Shows the model-indicated premium for each of the four coverages, highlighting their contributions to the total. These coverage indications remain unchanged when the proposed premium is adjusted.

From expected loss to the price we charge

Indicated rate = (pure premium including LAE + fixed expense per exposure) ÷ (1 – variable expense % – target profit %)

Non-Greek, Preferred-Plus

Sophomore · on campus · sprinklered · GPA 3.2 · Science

Component	Amount	% of premium
Losses and LAE	\$511.60	67.6%
Fixed underwriting expense	\$51.54	6.8%
Variable underwriting expense	\$155.92	20.6%
Underwriting profit	\$37.85	5.0%
Premium charged	\$756.90	100.0%

Greek, Non-Standard

Grad student · off campus 4.0 mi · not sprinklered · GPA 2.8 · Business

Component	Amount	% of premium
Losses and LAE	\$4,192.35	73.5%
Fixed underwriting expense	\$51.54	0.9%
Variable underwriting expense	\$1,175.05	20.6%
Underwriting profit	\$285.21	5.0%
Premium charged	\$5,704.15	100.0%

The fixed expense is the same \$51.54 for both students, but it is 6.8% of one premium and 0.9% of the other.

We set it by averaging pure premium including LAE across the book and multiplying by the NAIC general expense estimate of 5.1%.

How we validated the model

Balance check

Modelled pure premium against actual losses, by coverage, by program and by split

Book total 1.038 training · 1.109 holdout

Coverage	Non-Greek training	Non-Greek holdout	Greek training	Greek holdout
Personal Property	1.005	0.936	0.980	1.306
Additional Living Expense	1.044	1.126	0.939	0.934
Guest Medical	1.132	1.717	1.050	1.015
Liability	1.256	1.014	0.758	0.536
Total policy	1.100	1.314	0.978	0.956

Each figure is modelled pure premium divided by actual losses. A value of 1.00 is exact balance.

The Liability model is built on only 85 claims, so those ratios need more data before they carry weight. Fitting separate models by program is the next step we would take.

Year one pro forma

Line	Basis	Amount	% of premium
Written premium	10,000 policies × \$1,427.47	\$14,274,700	100.0%
Losses and LAE	modelled pure premium × 1.1284 LAE load	\$10,105,000	70.8%
Fixed underwriting expense	10,000 policies × \$51.54	\$515,400	3.6%
Variable underwriting expense	20.6% of premium, NAIC 2024	\$2,940,588	20.6%
Underwriting profit	5.0% of premium, target	\$713,712	5.0%
Combined ratio	losses and expenses ÷ premium	\$13,560,988	95.0%

Assumes full retention at indicated rates. The three-year view on the next slide relaxes that.

What the book looks like in three years

Base case

Full retention, trend applied by coverage

Combined ratio holds near 95%

Adverse selection

Greek share falls from 19.6% toward single digits

Loss ratio falls, expense ratio rises, combined ratio unchanged

Growth

Non-Greek volume grows on a competitive price

Fixed costs spread further, margin improves

The insight to land

Greek members are 19.6% of policies but 46.6% of premium. If they leave, the combined ratio stays at 95.0%, because every policy is priced to it. What disappears is scale: about \$333,000 of the \$714,000 underwriting profit. The segment is not a margin problem, it is a volume problem, which is why the answer is to price it rather than decline it.

What we recommend

1

Adopt the two-program, four-tier plan

Every policyholder gets a price built from their own risk, not the book average.

2

Price Greek and Non-Greek separately

Two programs, each with its own base rates and factor tables, so the difference in loss experience is carried by the rate rather than by the rest of the book.

3

Differentiate Business and Social Greek-life

Greek membership is our largest rating variable at four to five times, but it is a single binary flag covering organizations with very different risk profiles.

4

Hold Liability provisional and monitor

Our Liability model is built on only 85 claims, which is too few to price confidently. Refit it as more claims come in

Where each case requirement is answered

Requirement	Slide	Requirement	Slide
1a Is the data appropriate	3, 4	1b Additional data sources	22
1c Transformations and cleansing	4	2a Modeling techniques and assumptions	5, 6, 7
2b Variable selection	8, 9	2c Appropriate for new policyholders	17, 20
2d Evaluation graphs	17	3a Satisfaction, renewal, new vs renewal cost	22
3b Credits and debits within tier	11, 14	3c Expense component of the rate	16
3d Rating strategy over time	19	4a How to use the model output	11, 13, 14, 15
4b Population and profitability, 1 to 3 years	19	4c Pro forma and business plan	18, 20

Slide numbers match the position in the deck. Re-check them after any slide is added or removed.

Assumptions

Data and exposure

- Exposure is exactly one policy-year per student, so no offset is used.
- No deductible, coinsurance or limit censoring, verified rather than assumed.
- Claim counts are independent of claim sizes, so $E[S] = E[N] \cdot E[X]$
- Claims within a policyholder are independent and identically distributed.

Expense, inference and scope

- LAE is proportional to loss dollars at 12.84%, the NAIC Homeowners ratio.
- Variable expense 20.6%, fixed expense 5.1% of premium, both from NAIC 2024.
- Target profit and contingencies of 5.0%, the filed convention. No investment income offset taken.
- Rating variables are close to independently assigned in this data

Code Links

GitHub

[GitHub](#)